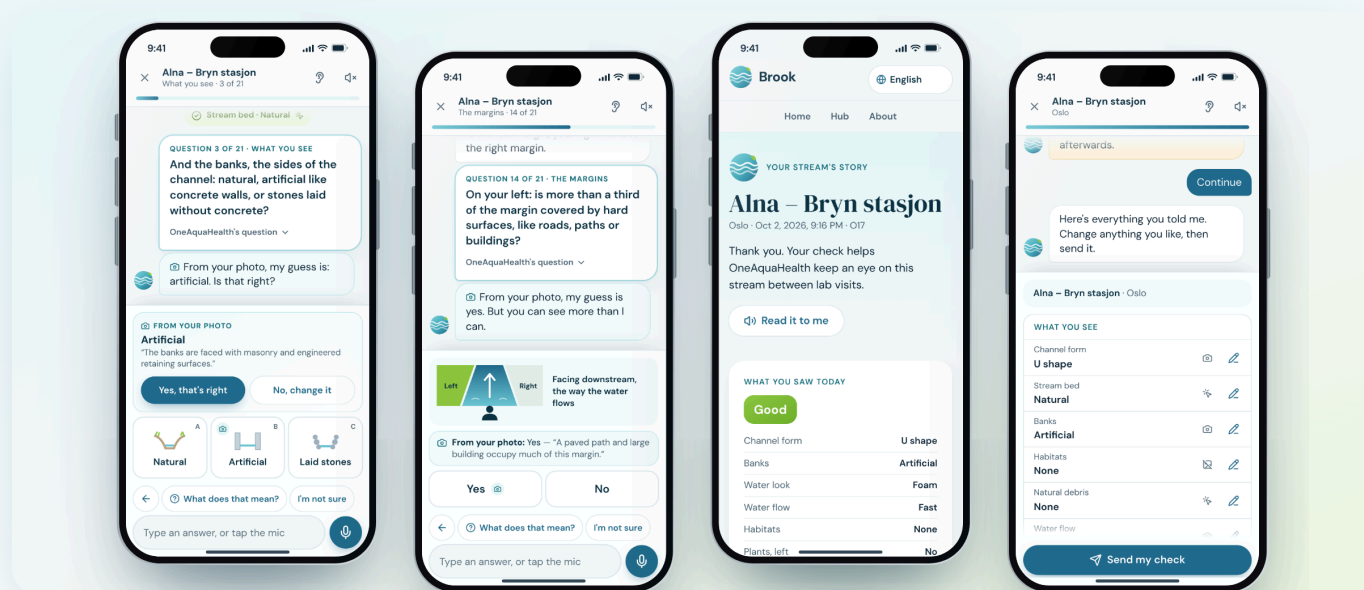


Talk to your stream.

A friendly guide for OneAquaHealth's citizen stream check. Talk or tap in seven languages, get every hard word explained, and send data scientists can use right away.



TRY IT IN ONE MINUTE

brook-oah.vercel.app/check?demo=1

VIDEO, 3 MINUTES

youtube.com/watch?v=fHfNSAoPz7I

CODE, OPEN SOURCE

github.com/kinnuworks/brook

The problem

At the stream, volunteers meet questions like *"Is more than one third of the left margin covered by impervious areas?"* Left of what? What does impervious mean? People guess or give up. The Track 1 brief names it: *"complex tools, confusing terminology, and low participation."*

What Brook does

- Asks each official question in plain words, by voice or tap, and explains any word.
- Suggests answers from your photos; nothing is saved until you confirm.
- Takes a gentle second look when two answers don't fit. You always decide.
- Shows your stream's story: OneAquaHealth's lab results next to what you saw.
- Sends OneAquaHealth's exact format, plus HL7 FHIR health data.

0 errors

official HL7 validator, against OneAquaHealth's FHIR guide

93%

of 238 realistic replies in 7 languages understood without AI

132 / 134

photo suggestions right, on 36 stream photos never seen before

0 problems

accessibility scan (WCAG 2.1 AA), every screen

How Brook meets the judging criteria

Impact & alignment with OneAquaHealth

30%

- Built on OneAquaHealth's own questions, answer codes, 106 research sites, lab results and FHIR guide.
- Sends checks in the exact format OneAquaHealth's app already uses, so they can be used unchanged.
- Your stream's story: OneAquaHealth's lab results for that site next to what you saw, with One Health tips for people, pets and the stream.
- Works in the seven languages of OneAquaHealth's cities.

Innovation & creativity

20%

- A conversation instead of a form: talk or tap, and ask "what does that mean?" at any time.
- Photos suggest answers; the person confirms every one. A second look when two answers don't fit.
- For researchers: which questions confuse volunteers, so the form can be improved. No paper form can show this.

Technical implementation

20%

- Health data checked with the official HL7 validator against OneAquaHealth's FHIR guide: 0 errors; sent to their HL7 Europe sandbox.
- Understands 93% of 238 realistic replies in seven languages with no AI; photo suggestions right 132 times out of 134 on 36 unseen stream photos.
- 111 automatic tests, a robot walkthrough of a full check, an offline test (kept and sent when signal returns).
- AI limited to each question's own answers, with spending limits; about 0.1 cent per check, measured.

Usability & user experience

15%

- Plain words, small drawings, and a "face downstream" diagram for the left/right questions.
- Voice or tap; a check takes about five minutes; works on phones and laptops.
- Accessibility scan (WCAG 2.1 AA): 0 problems on every screen.

Feasibility & scalability

15%

- €0 to host at pilot scale; about €15 of AI for 10,000 checks a year, and €0 with AI switched off.
- No app store and no accounts; open source (MIT); OneAquaHealth's websites can embed it.
- Data stays in the EU; photos have their location removed on the phone.

What we don't claim

- Brook hasn't been tried by volunteers at a real stream yet; a pilot with OneAquaHealth's citizen-science teams is the next step.
- The research hub starts with made-up example checks, clearly labelled "Simulated" and never sent to OneAquaHealth.
- The seven languages haven't yet been reviewed by native speakers.